Comets, meteors, and asteroids

Billions of comets and asteroids are invisible to us on Earth because they are too distant or too small to be seen. Yet some comets appear to grow and then make a spectacular show in the night sky. On any given night, meteors produced by cometary dust flash into view, and asteroids can crash-land on Earth's surface.

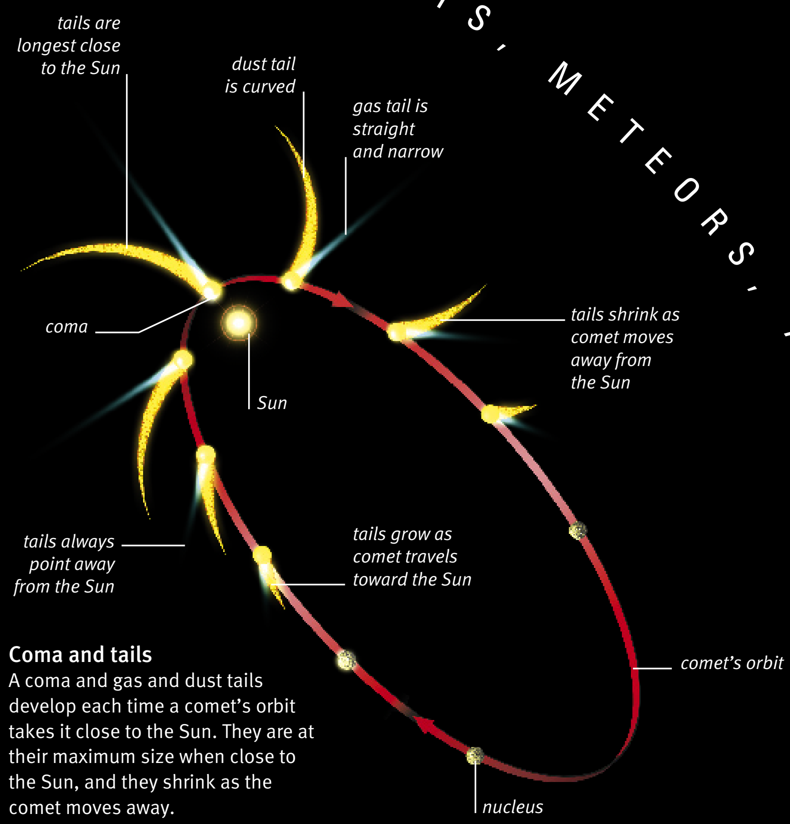
COMETS

Comets are often referred to as dirty snowballs. But these cosmic snowballs are not ball-shaped, and they are on a giant scale. Comets are irregularly shaped, city-sized lumps of snow and rocky dust that follow orbits around the Sun. More than a trillion of them together make up the vast Oort cloud, where they have been since the birth of the solar system 4.6 billion years ago.

Comets only become visible when they leave the Oort cloud and travel in toward the Sun. Then, the snowball, called a nucleus, is warmed by the Sun's heat. The snow is converted into gas, and this and loosened dust jet away from the nucleus. When closer to the Sun than Mars, the nucleus is surrounded by a cloud of gas and dust, known as a coma, and it develops two tails.

The comet is now large enough and bright enough to be seen from Earth. Typically, two or three a year are seen as a fuzzy patch of light through binoculars. Three or four a century, such as Comet McNaught, are truly impressive and easily visible to the naked eye.

More than 2,000 comets have been recorded within the vicinity of the Sun, and about 200 are periodic. Most are named after their discoverers.

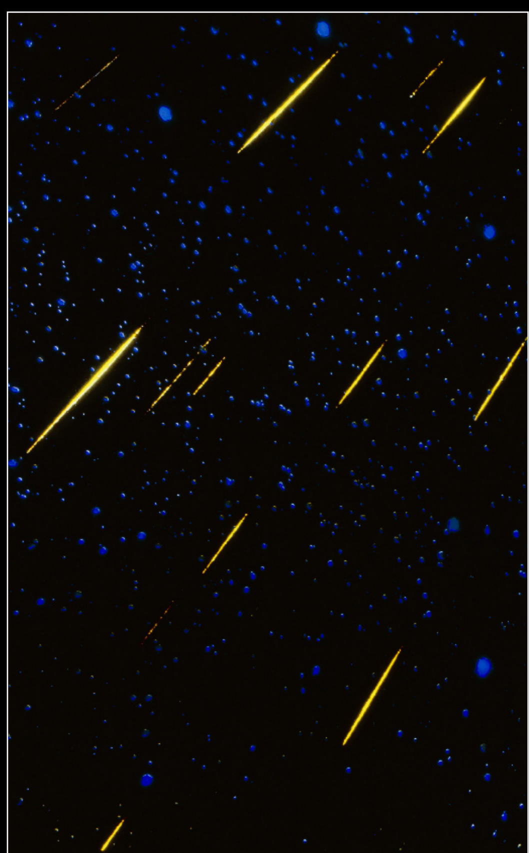


Coma and tails
A coma and gas and dust tails develop each time a comet's orbit takes it close to the Sun. They are at their maximum size when close to the Sun, and they shrink as the comet moves away.

METEORS

On any night of the year it is possible to see a meteor—a short-lived trail of light in the sky. These transient flashes, popularly called shooting stars, are produced by small fragments of a comet or, sometimes, an asteroid. As the fragment, or meteoroid, moves through Earth's atmosphere, it produces a trail of excited atoms, which in turn produce light. This streak of light is the meteor. It lasts for less than a second and has an average brightness of magnitude 2.5.

The best time to see a meteor is during the hours before dawn, when you'll be on the side of Earth facing the incoming meteoroids. Although meteors appear every night, they are not predictable. To maximize your chances of seeing one, it is best to observe when a meteor shower is expected. These happen on the same dates every year and are produced as Earth plows into a stream of comet particles. Shower details are in the Monthly Sky Guide (see pp.44–69).



Leonid shower
The Leonid meteor shower occurs in mid-November. As Earth orbits, it passes through a stream of particles lost by Comet Tempel–Tuttle. The dust particles produce meteors that appear to radiate from a point in the constellation of Leo.

ASTEROIDS

Asteroids are dry, dusty lumps of rock that failed to make a rocky planet. Most are irregular in shape, and just a handful over about 200 miles (320 km) across are round. They range in size from Ceres at 596 miles (960 km), through boulders and pebbles to dust-sized particles. There are 100,000 larger than about 12.5 miles (20 km) across. As size diminishes, the quantity increases. Most orbit the Sun between Mars and Jupiter.

Ida and Dactyl
The asteroid Ida is 37 miles (60 km) long and shows signs of past collisions. It journeys around the Sun every 4.8 years. Its tiny moon, Dactyl, just 1 mile (1.6 km) long, orbits Ida every 27 hours.

Meteorites
When an asteroid makes it through Earth's atmosphere and lands on the surface, it is called a meteorite. More than 22,500 have been cataloged. This one has a dark crust that formed as it fell through the atmosphere.